# Investment Portfolio Proposal

Strategic asset allocation and risk analysis for a private-banking client

**Client:** Mr. Omar Haddad (fictional), Dubai, United Arab Emirates
**Mandate:** USD 500,000, advisory, moderate risk profile
**Investment horizon:** 10 years, with annual income of USD 15,000
**Market data:** Index and yield data to 13 October 2026; ETF closing prices to 12 October 2026
**Date:** 14 October 2026
**Prepared by:** Yamen Agha

This document is a case study prepared for educational and portfolio purposes. The client is fictional. Market data are real and sourced as listed in Appendix A. Nothing in this document is investment, tax or legal advice. See Section 8.

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## Executive summary

Mr. Haddad is a 45-year-old Syrian engineer living in Dubai who wants to invest USD 500,000 for ten years. He needs about USD 15,000 a year from the portfolio, prefers liquid US-dollar assets, and is open to a Sharia-compliant version. His risk profile is moderate. This proposal recommends a diversified portfolio of ten exchange-traded funds: **45% equities, 37% bonds, 10% real assets and 8% cash**. Its forward-looking expected return is 6.1% a year before advisory fees, with expected volatility of about 8.3%.

The allocation takes account of today's market. The 10-year US Treasury yield is 5.21% and the 10-year real (inflation-protected) yield is 2.88%, the highest levels since 2007 and 2008. High yields make high-quality bonds useful again, both for income and as a buffer against equity losses, so the portfolio holds more fixed income than a typical moderate mandate and less US equity than a market-weighted portfolio. Gold and short-dated Treasury bills protect against the geopolitical and oil-price risks of the US-Iran conflict. The Gulf equity allocation is kept small (5%) because the client's salary already depends on the Gulf economy.

**Table 1. Recommended allocation at a glance**

| Asset class | Weight | USD | Funds |
|---|---|---|---|
| Equities | 45% | 225,000 | VTI, VEA, VWO, KSA, UAE |
| Fixed income | 37% | 185,000 | BND, SCHP |
| Real assets | 10% | 50,000 | IAU (gold), VNQ (REITs) |
| Cash (T-bills) | 8% | 40,000 | BIL |
| **Total** | **100%** | **500,000** | **Weighted expense ratio 0.092%** |

**What the history shows.** We simulated the allocation over the ten years from 30 September 2016 to 30 September 2026, with quarterly drift checks and trading costs. It returned 7.5% a year with volatility of 8.3%. Its worst fall was −17.9%, in 2022. A US 60/40 portfolio returned 9.7% a year with volatility of 10.2% and a worst fall of −21.0%. The global balanced benchmark (AOR) returned 8.0% with volatility of 10.0%. The proposal gave up some return against the 60/40, mostly because US large-cap stocks were exceptionally strong over the decade. In exchange, it lost less in every major sell-off before 2026. In the COVID crash it fell −17.2%, against −21.0% for the 60/40. In calendar 2022 it fell −13.0%, against −16.0%.

**Risk and income.** On a one-year view, the 95% Value at Risk is about 7.6% of the portfolio (USD 38,027). We ran 10,000 simulated ten-year paths using our forward-looking assumptions, with inflation-linked withdrawals of USD 15,000 a year and a 0.50% annual fee. The median value after ten years was USD 628,818, and the portfolio was never used up. The chance of ending above the initial USD 500,000 in nominal terms was 76%. In real terms, after inflation, the chance was 50%. A 3% withdrawal rate is therefore sustainable, but it uses up most of the expected real return. If preserving purchasing power matters most, the client should withdraw closer to 2.5% or accept more equity risk.

**Implementation.** As a non-US person, the client would face US estate-tax exposure on US-domiciled funds above a USD 60,000 filing threshold. He would also pay 30% US withholding tax on equity dividends. We therefore recommend holding the portfolio through equivalent Irish-domiciled (UCITS) funds where possible, as listed in Section 7. The US-listed funds are used in the analysis because they have longer price histories.

# 1. Client profile and investment policy statement

## 1.1 Background

Mr. Omar Haddad (fictional) is 45 and a Syrian national. He has lived in Dubai for 15 years, where he works as a senior project engineer for an energy-infrastructure contractor. He is married with two children, aged 12 and 15. Each year he sends money to his parents in Aleppo. He has built up USD 500,000 in liquid savings, now held in deposits at a UAE bank, and wants a professionally structured portfolio. He has a separate emergency reserve of six months' expenses in UAE dirham deposits. He also has an end-of-service gratuity entitlement through his employer, and an education savings plan for his children. Neither forms part of this mandate. He rents his home and has no debts.

He has invested before through a bank-sold balanced fund. He stayed invested through the 2020 crash, but sold part of his holdings during the 2022 fall, which he now regrets. He prefers simple, liquid instruments priced in US dollars. Because the UAE dirham is pegged to the US dollar, his spending currency and his investment currency are effectively the same. He has asked to see a Sharia-compliant version of the portfolio and will decide between the two after this review.

## 1.2 Investment objectives

**Return objective.** The portfolio should earn a long-run total return of about 6% a year before fees, roughly inflation plus 3.5%. That would fund withdrawals of USD 15,000 a year (3% of the starting value), increased each year in line with inflation, and broadly keep the capital's purchasing power over ten years. On the current 10-year breakeven inflation rate of 2.33%, keeping real capital intact requires a return of about 5.3% a year after fees, or about 5.8% before the assumed 0.50% fee. That is close to the portfolio's expected return, so the objective is achievable but not certain.

**Risk objective.** Expected annual volatility should be between 7% and 10%. The one-year 95% Value at Risk should be no more than 10% of the portfolio. The client should not normally face a peak-to-trough fall of more than about 20%. That is the level at which, by his own account, he would be tempted to sell.

## 1.3 Risk tolerance assessment

We assessed risk tolerance in two parts. The first is willingness: how the client feels about losses, measured with a short questionnaire. The second is ability: how much loss his finances can absorb. Table 2 shows the questionnaire. Each answer scores 1 (most cautious) to 5 (most aggressive).

**Table 2. Risk tolerance questionnaire results**

| Question | Client answer | Score (1-5) |
|---|---|---|
| Primary goal for this money | Growth with some income | 3 |
| Reaction to a 20% portfolio fall | Hold, but would feel uncomfortable | 3 |
| Past behaviour in a downturn (2022) | Sold part of holdings | 2 |
| Investment experience | Funds and deposits; some ETF knowledge | 3 |
| Time before most of the money is needed | More than 10 years | 4 |
| Share of liquid net worth in this portfolio | About 70% | 2 |
| Stability of employment income | Stable, senior role, gratuity accrual | 4 |
| Preferred trade-off | Moderate return, moderate swings | 3 |
| **Total** | | **24 of 40** |

Scoring bands: 8-16 conservative; 17-24 moderately conservative; 25-32 moderate; 33-40 growth.

A score of 24 puts the client in the lower half of the moderate band. His ability to take risk is above average. His income is stable, the withdrawal need is modest at 3% a year, the horizon is long, and he keeps a separate emergency reserve. However, his job depends on the Gulf energy sector, and his UAE residence is tied to his employment. Both risks would hit at the same time in a regional downturn, so we do not let his

higher ability override his willingness. When the two differ, the IPS follows the lower of the two. The overall risk tolerance is **moderate**.

## 1.4 Constraints

**Time horizon.** Ten years for the strategic plan, reviewed each year. The client will be 55 at the end of the period and may then move to a retirement-income stage. The horizon is long enough for equity risk, but it is one period, not several, and withdrawals start straight away.

**Liquidity.** Withdrawals of USD 1,250 a month, increased each year for inflation. The client also wants to be able to raise up to 10% of the portfolio within three business days for family needs. All holdings are daily-traded ETFs. The 8% T-bill sleeve covers about 2.7 years of withdrawals, so the portfolio never needs to sell equities in a falling market to pay income.

**Tax.** The UAE does not currently levy personal income tax on individuals' investment income. US-domiciled funds, however, are subject to US rules for non-resident aliens. There is no US income tax treaty with the UAE (IRS treaty list), so dividends paid by US-domiciled equity funds face 30% US withholding tax. Distributions that US bond funds designate as interest-related dividends are generally exempt. More importantly, US-domiciled shares and ETFs count as US-situs assets for US estate tax. The executor of a non-resident non-citizen must file Form 706-NA if those assets exceed USD 60,000. Irish-domiciled UCITS funds largely avoid both problems. The client should confirm his tax residence position, including any Syrian obligations, with a qualified tax adviser.

**Legal and regulatory.** The account will be opened with a regulated private bank in the UAE, which will carry out full know-your-customer checks, including source-of-wealth documentation (salary records and bank statements). The client's tax-residence information will be reported under the OECD Common Reporting Standard. Transfers to family in Syria must use licensed channels and pass the bank's sanctions and anti-money-laundering screening. The rules on Syria have changed substantially since 2025, so the bank's compliance team should confirm the current position before any regular transfer. For succession, the client should take legal advice on a registered UAE will. Without one, Sharia-based inheritance rules may apply to his assets by default.

**Unique circumstances.** (1) His human capital (future salary) is concentrated in the Gulf energy economy, so we limit Gulf equities. (2) His residence depends on his employment, so his assets need to be portable and held with an internationally connected custodian. (3) He sends money regularly to family in Syria. (4) He wants to be able to adopt a Sharia-compliant version of the portfolio. (5) He sold during the 2022 decline. A clear rebalancing rule and a cash reserve are designed to stop that from happening again.

## 1.5 Policy summary

**Table 3. Strategic policy weights and ranges**

| Asset class | Strategic weight | Permitted range | Main purpose |
|---|---|---|---|
| Equities | 45% | 40% - 60% | Long-term real growth |
| of which Gulf (KSA, UAE) | 5% | 0% - 6% | Regional opportunity, capped for human-capital overlap |
| Fixed income (aggregate, TIPS) | 37% | 25% - 45% | Income, ballast, inflation protection |
| Gold | 7% | 0% - 10% | Geopolitical and inflation hedge |
| REITs | 3% | 0% - 8% | Real-asset income |
| Cash (T-bills) | 8% | 3% - 15% | Withdrawal reserve, liquidity |

Benchmarks: the policy is judged against (a) its own strategic weights, rebalanced under the same rules, (b) a US 60/40 portfolio (60% SPY, 40% AGG, rebalanced quarterly) and (c) the iShares Core 60/40 Balanced Allocation ETF (AOR), a global balanced fund used as an investable peer benchmark.

# 2. Market outlook

This section reflects market data to the close on 13 October 2026. ETF closing prices run to 12 October 2026, the latest daily close the price feed had published when the data were retrieved. The US consumer price report for September is due at 08:30 New York time on 14 October 2026, after this proposal was finalised. It may move rates and should be checked before implementation.

**Table 4. Market snapshot**

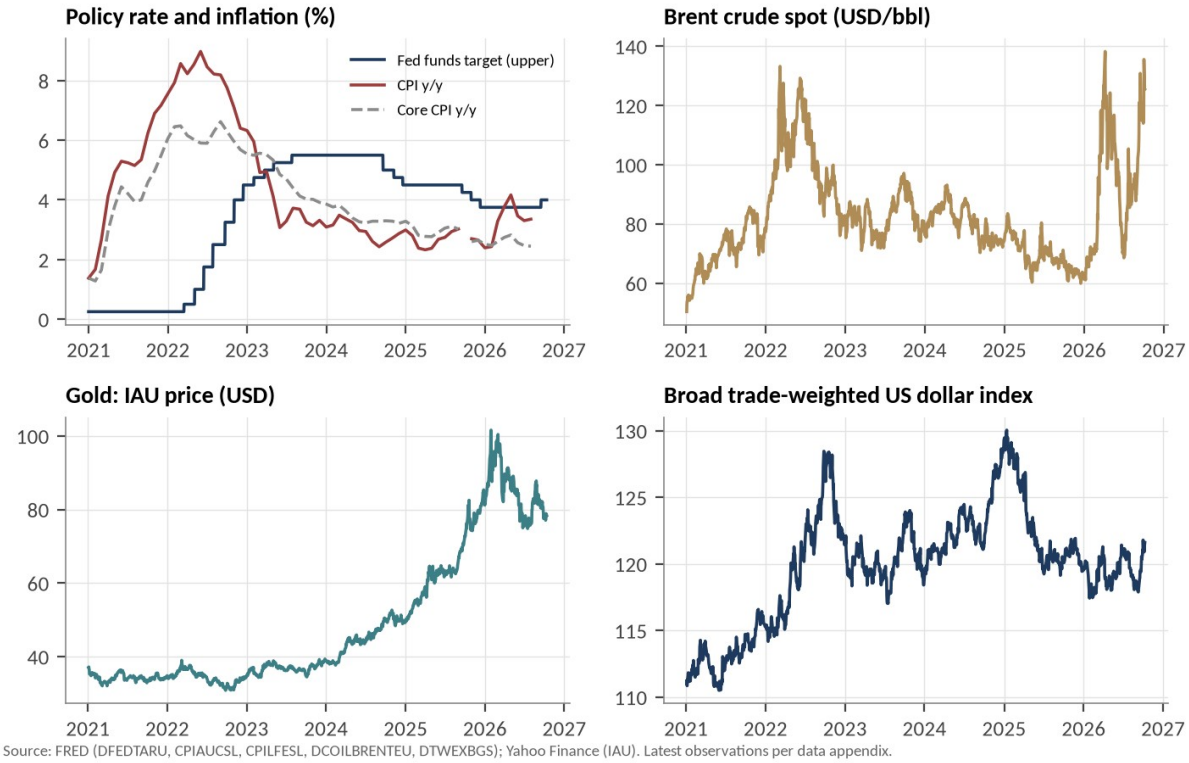
| Indicator | Latest | Context | Source and date |
|---|---|---|---|
| Fed funds target range | 3.75% - 4.00% | Raised 25 bp on 16 Sep 2026 | Federal Reserve, 16 Sep 2026 |
| US CPI inflation, y/y (core) | 3.4% (2.4%) | August 2026 data | BLS, 11 Sep 2026 |
| 3-month Treasury bill | 4.26% | 4.02% a year earlier | US Treasury, 13 Oct 2026 |
| 2-year Treasury | 4.79% | 3.48% a year earlier | US Treasury, 13 Oct 2026 |
| 10-year Treasury | 5.21% | 4.03% a year earlier | US Treasury, 13 Oct 2026 |
| 10-year TIPS real yield | 2.88% | Highest since 2008 | US Treasury, 13 Oct 2026 |
| 10-year breakeven inflation | 2.33% | Market-implied inflation | FRED T10YIE, 13 Oct 2026 |
| S&P 500 | 7,805.45 | +14.0% year to date | AP / Yahoo Finance, 13 Oct 2026 |
| VIX | 15.27 | Implied equity volatility | Yahoo Finance, 13 Oct 2026 |
| Brent crude (futures settle) | USD 105.69 | Up more than 70% in 2026 | Reuters, 13 Oct 2026 |
| Gold (COMEX front month) | USD 4,181 | −3.7% YTD; about 25% below Jan record | Yahoo Finance, 13 Oct 2026 |
| US Dollar Index (DXY) | 102.17 | +4.0% year to date | Yahoo Finance, 13 Oct 2026 |
| US investment-grade spread (OAS) | 0.83% | High yield: 3.11% | FRED (ICE BofA), 12 Oct 2026 |
| US unemployment rate | 4.2% | September 2026 | FRED (BLS) |

## 2.1 Growth, inflation and policy

The main macro story of 2026 is an energy shock on top of a resilient US economy. The US-Israeli war with Iran began with strikes in late February and is now in its eighth month. It has disrupted shipping near the Strait of Hormuz and pushed oil prices sharply higher. Brent futures settled at USD 105.69 a barrel on 13 October, after a close of USD 106.65 the day before, the highest since mid-September (Reuters). The US Energy Information Administration's dated Brent spot assessment, the price for prompt physical delivery, has been higher still, between about USD 114 and USD 136 a barrel between 21 September and 6 October (FRED). That points to a tight physical market.

Inflation has picked up as a result. US CPI rose 3.4% in the year to August, with energy prices up 16.3%. Core inflation, which excludes food and energy, was a more moderate 2.4% (BLS). With growth still solid and inflation above target, the Federal Reserve under Chair Warsh raised its policy rate by 25 basis points to 3.75%-4.00% on 16 September, a unanimous decision. The Committee's median projections show PCE inflation of 3.7% in 2026, falling to 2.3% in 2027, and real GDP growth of 2.3% this year (Federal Reserve). The IMF's October World Economic Outlook, published on 13 October, projects world growth of 3.1% in 2026 and 3.4% in 2027, global headline inflation of 4.7% this year, and a 0.3% contraction for the Middle East and Central Asia region. It describes the global economy's resilience as "reaching its limits".

**Figure 1. Policy rate, inflation, oil, gold and the US dollar since 2021**

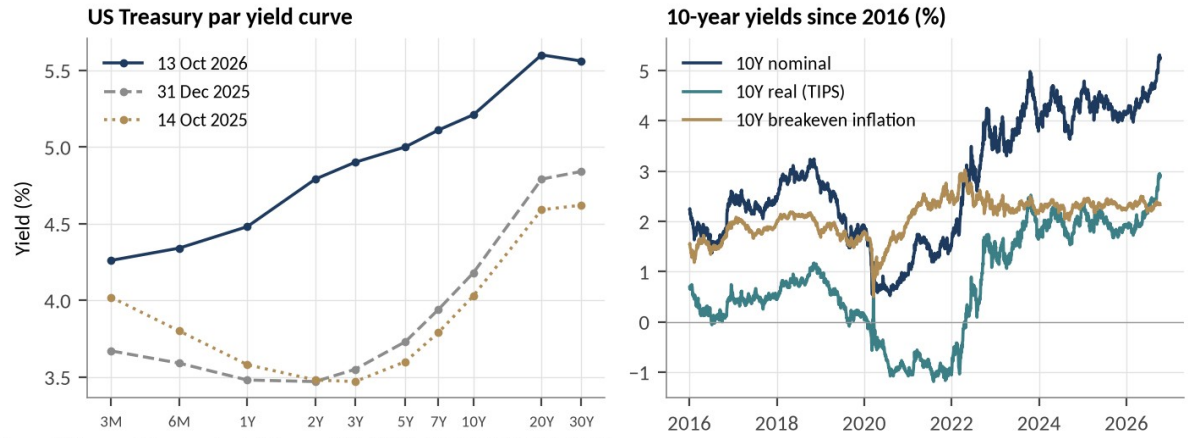



Source: FRED (DFEDTARU, CPIAUCSL, CPILFESL, DCOILBRENTEU, DTWEXBGS); Yahoo Finance (IAU). Latest observations per data appendix.

## 2.2 Interest rates

Yields have risen sharply. The 10-year Treasury yield ended 13 October at 5.21%, up about 1.2 percentage points from a year earlier. Its 52-week high was 5.31% on 5 October (Dow Jones). The last time the 10-year yield was this high was June 2007. The 10-year real yield, at 2.88%, is the highest since late 2008. A year ago the yield curve was inverted at the short end; it now slopes upward from 4.26% at three months to 5.56% at 30 years. Investors are again being paid to hold longer-dated bonds. Credit spreads, by contrast, remain tight, with investment-grade bonds yielding only 0.83% more than Treasuries. We therefore prefer to take duration risk through government-heavy aggregate bonds and TIPS rather than to take credit risk.

**Figure 2. US Treasury yield curve and 10-year nominal, real and breakeven yields**




Source: US Treasury daily par and real yield curves (13 Oct 2026); FRED DGS10, DFII10, T10YIE.

## 2.3 Equities, gold and the dollar

US equities have looked past the conflict. The S&P 500 closed at 7,805.45 on 13 October, just below the record it set the previous week. It is up 14.0% for the year, and the VIX volatility index is a calm 15.3. With the 10-year Treasury yield above 5%, equities must now clear a much higher hurdle than for most of the past

15 years, when that yield averaged about 2.6%. That argues for a neutral, not aggressive, equity weight and for diversification outside US large caps. Gold peaked near USD 5,590 an ounce in late January, when fear of war was at its highest. It has since fallen about a quarter, to around USD 4,181, as real yields rose and the dollar firmed. The dollar index is up 4.0% this year. For a dollar-based investor with dirham spending, a stronger dollar is not a risk in itself, though it does reduce the dollar value of non-US assets.

### 2.4 Gulf markets

Gulf equity markets have not benefited from high oil prices, because the region is close to the fighting. Saudi Arabia's Tadawul index had erased its 2026 gains by early October. It closed at 10,360.74 on 11 October after a deadly attack on Riyadh's King Khalid International Airport, which Yemen's Houthi movement claimed. Qatar's market fell to its lowest close since May 2020 (Reuters via Zawya; 60 Arabia). The IMF expects the wider region's economy to shrink this year before recovering in 2027. Valuations may become attractive, but the near-term risk is high, and the client's own income is already exposed to the region.

### 2.5 Implications for the portfolio

- **Higher yields make bonds useful again.** Starting yields are the best predictor of a bond fund's return over its duration, so a 37% bond allocation now earns about 5% with much lower volatility than equities.
- **Inflation protection matters.** With headline inflation above 3% and an oil supply shock under way, TIPS (12%) and gold (7%) provide protection that ordinary bonds do not.
- **Keep equities moderate and diversified.** US valuations and the narrow equity premium argue against overweighting US stocks, so 56% of the equity sleeve is in the US, against roughly two-thirds in global market-cap indices.
- **Keep Gulf exposure small.** The conflict and the client's employment risk both point to a small, capped Gulf allocation.
- **Hold a cash reserve.** T-bills yield more than 4%, so an 8% reserve to fund withdrawals costs very little in expected return.

# 3. Strategic asset allocation

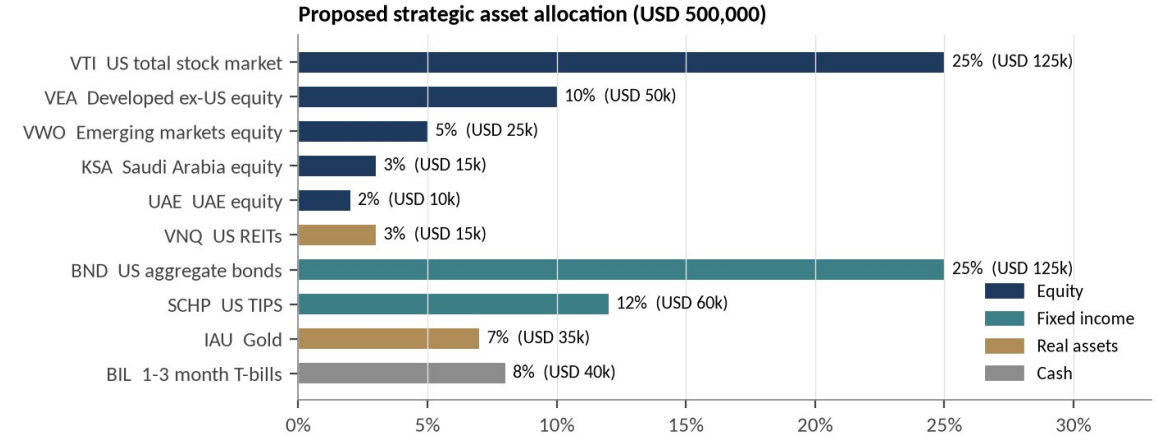
Table 5 sets out the recommended allocation. Each sleeve uses a large, low-cost, daily-traded ETF. The weighted expense ratio is 0.092% a year, or about USD 458 on USD 500,000. The weighted trailing distribution yield is 2.83%, or about USD 14,141 a year before withholding tax. Distributions therefore cover most of the planned USD 15,000 withdrawal. Shares are calculated at closing prices on 12 October 2026.

**Table 5. Recommended strategic asset allocation**

| Ticker | Fund | Class | Weight | USD | Expense ratio | Yield | Price | Shares |
|---|---|---|---|---|---|---|---|---|
| VTI | Vanguard Total Stock Market ETF | Equity | 25% | 125,000 | 0.03% | 1.04% | 381.57 | 327 |
| VEA | Vanguard FTSE Developed Markets | Equity | 10% | 50,000 | 0.03% | 2.43% | 69.95 | 714 |
| VWO | Vanguard FTSE Emerging Markets | Equity | 5% | 25,000 | 0.06% | 2.04% | 59.73 | 418 |
| KSA | iShares MSCI Saudi Arabia ETF | Equity | 3% | 15,000 | 0.75% | 2.89% | 35.66 | 420 |
| UAE | iShares MSCI UAE ETF | Equity | 2% | 10,000 | 0.59% | 4.39% | 19.31 | 517 |
| VNQ | Vanguard Real Estate | Real assets | 3% | 15,000 | 0.13% | 3.74% | 91.36 | 164 |
| BND | Vanguard Total Bond Market | Fixed income | 25% | 125,000 | 0.03% | 4.18% | 70.25 | 1,779 |
| SCHP | Schwab U.S. TIPS ETF | Fixed income | 12% | 60,000 | 0.03% | 4.99% | 25.21 | 2,380 |
| IAU | iShares Gold Trust | Real assets | 7% | 35,000 | 0.25% | n/a | 78.34 | 446 |
| BIL | SPDR Bloomberg 1-3 Month T-Bill ETF | Cash | 8% | 40,000 | 0.14% | 3.66% | 91.50 | 437 |
| **Total** | | | **100%** | **500,000** | **0.092%** | **2.83%** | | |

Expense ratios and trailing 12-month distribution yields from stockanalysis.com, cross-checked against Yahoo Finance, retrieved 14 Oct 2026. Prices are closing prices on 12 October 2026 (Yahoo Finance).

**Figure 3. Proposed allocation by fund**




Weights as proposed in Section 3; USD amounts based on an initial portfolio of USD 500,000.

## 3.1 Rationale by sleeve

**US equities, VTI (25%).** The total US market (more than 3,500 stocks) at a 0.03% fee is the core growth engine. It is held below its global market weight because US valuations are demanding at a time of 5% bond yields.

**Developed ex-US equities, VEA (10%), and emerging markets, VWO (5%).** These reduce the portfolio's dependence on one market and one sector (US technology). They also add higher dividend yields: 2.4% and 2.0%, against 1.0% for VTI.

**Gulf equities, KSA (3%) and UAE (2%).** These give a regional allocation the client understands and some exposure to Gulf capital-market reforms. They are capped at 6% in total because the client's salary and residence already depend on the region, and because the conflict raises near-term risk. Both funds are expensive, at 0.75% and 0.59%, so the cap also limits cost.

**US aggregate bonds, BND (25%).** A broad, mostly government and agency bond portfolio of intermediate duration. At current yields it is the main source of income, and it is the asset most likely to rise in a recession.

**US TIPS, SCHP (12%).** Inflation-linked Treasuries lock in a real yield of about 2.7%-2.9% plus realised inflation. This directly protects the purchasing power of the client's withdrawals.

**Gold, IAU (7%).** Gold has a low correlation with equities and tends to hold up in geopolitical and currency stress. Its correlation with US stocks over the past ten years was 0.12. It pays no income, so the weight is modest.

**US REITs, VNQ (3%), and T-bills, BIL (8%).** A small listed real-estate sleeve adds property income. The T-bill sleeve is the withdrawal reserve and the source of cash for rebalancing.

## 3.2 Capital market assumptions

The forward-looking assumptions in Table 6 are built from current market yields as of 13 October 2026. Cash earns the 3-month bill yield. Aggregate bonds earn the 5-year Treasury yield, which roughly matches BND's duration, plus a 0.2% spread allowance. TIPS earn the 5-year real yield plus 10-year breakeven inflation. Gold earns inflation plus 1%. Equity returns are the cash yield plus an equity risk premium of 2.5%-4.0%. The equity premiums are our own judgement and are deliberately conservative, given current valuations. Volatilities come from ten years of monthly returns. These are planning assumptions, not forecasts.

**Table 6. Capital market assumptions (annual, nominal USD)**

| Ticker | Asset | Forward return | 10y hist. mean | Volatility | Construction |
|---|---|---|---|---|---|
| VTI | US total stock market | 7.26% | 15.03% | 15.8% | 3M yield + 3.0% premium |
| VEA | Developed ex-US equity | 7.76% | 10.53% | 15.5% | 3M yield + 3.5% premium |
| VWO | Emerging markets equity | 8.26% | 8.53% | 15.5% | 3M yield + 4.0% premium |
| KSA | Saudi Arabia equity | 8.26% | 9.66% | 18.0% | 3M yield + 4.0% premium |
| UAE | UAE equity | 8.26% | 7.32% | 17.6% | 3M yield + 4.0% premium |
| VNQ | US REITs | 6.76% | 5.90% | 17.7% | 3M yield + 2.5% premium |
| BND | US aggregate bonds | 5.20% | 1.31% | 5.2% | 5Y yield + 0.2% |
| SCHP | US TIPS | 5.00% | 2.18% | 5.0% | 5Y real + breakeven |
| IAU | Gold | 3.33% | 12.46% | 15.5% | Breakeven + 1% |
| BIL | 1-3 month T-bills | 4.26% | 2.27% | 0.6% | 3M yield |

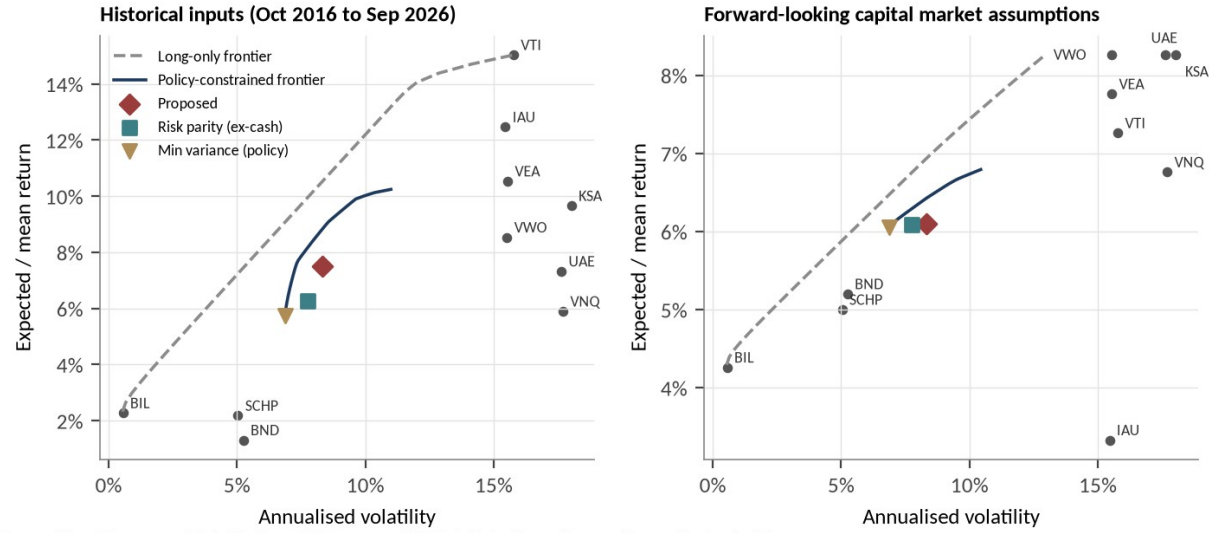
Yields: US Treasury par and real curves, 13 Oct 2026; breakeven: FRED T10YIE, 13 Oct 2026. Historical mean and volatility: monthly total returns Oct 2016 to Sep 2026.

The comparison with history is instructive. Over the past decade bonds returned only 1%-2% a year, because yields started near record lows and then rose sharply. US stocks and gold returned 12%-15%. Today's starting point is close to the reverse: bond yields are high, and equities and gold are expensive after strong runs. An allocation chosen by looking only at the past ten years would hold too much of yesterday's winners.

## 4. Portfolio optimisation: mean-variance and risk parity

We tested the proposed weights against two quantitative methods. **Mean-variance optimisation** (Markowitz) finds the portfolios with the highest expected return for each level of volatility, which together form the efficient frontier. **Risk parity** chooses weights so that each asset contributes equally to total portfolio risk, without needing return forecasts. Both use the covariance matrix of 120 monthly total returns for the ten funds, from October 2016 to September 2026. We ran the optimisation twice: once with long-only weights (0%-100% per fund), and once with the policy bounds of the IPS (for example, VTI 15%-40%, BND 10%-40%, gold at most 10%, equities 40%-60% in total, Gulf at most 6%).

**Figure 4. Efficient frontiers using historical and forward-looking expected returns**




Source: Yahoo Finance monthly total returns; US Treasury and FRED yields for forward assumptions; author's calculations.

**Table 7. Optimised portfolios compared with the proposal (ex-ante, annualised)**

| Portfolio | Equity | Real assets | Bonds | Cash | Volatility | Hist. return | Hist. Sharpe | Fwd return | Fwd Sharpe |
|---|---|---|---|---|---|---|---|---|---|
| Proposed | 45% | 10% | 37% | 8% | 8.3% | 7.5% | 0.60 | 6.1% | 0.22 |
| Max Sharpe, historical, policy bounds | 54% | 10% | 21% | 15% | 9.0% | 9.4% | 0.77 | 6.1% | 0.20 |
| Max Sharpe, historical, long-only | 58% | 42% | 0% | 0% | 11.4% | 13.6% | 0.97 | 5.7% | 0.12 |
| Max Sharpe, forward, policy bounds | 40% | 0% | 45% | 15% | 6.9% | 5.6% | 0.45 | 6.1% | 0.26 |
| Minimum variance, policy bounds | 40% | 0% | 45% | 15% | 6.9% | 5.8% | 0.48 | 6.1% | 0.26 |
| Risk parity (excluding cash) | 37% | 17% | 46% | 0% | 7.8% | 6.3% | 0.49 | 6.1% | 0.24 |

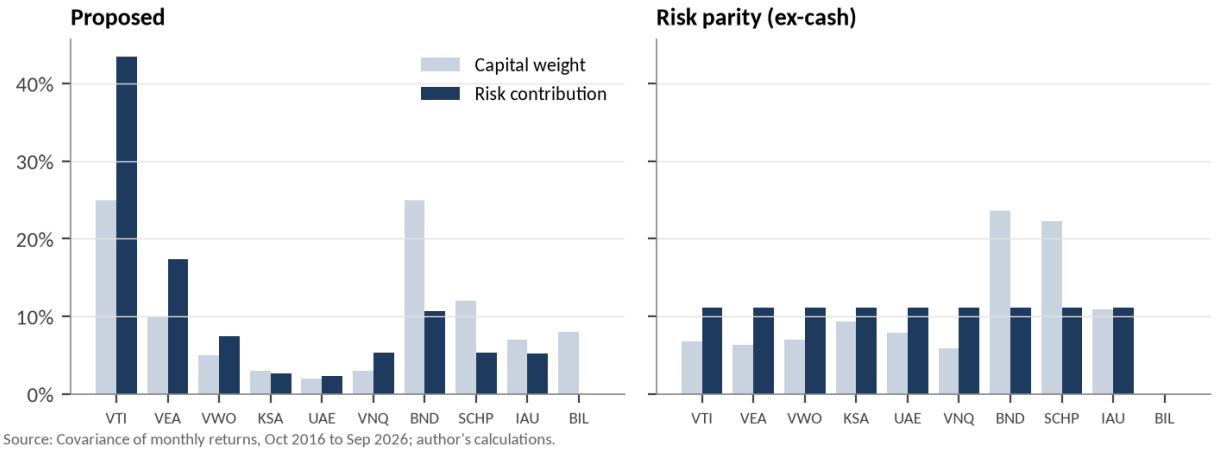
Historical Sharpe uses the 10-year average T-bill rate (2.49%); forward Sharpe uses the current 3-month yield (4.26%). Full weights are in results/optimised_weights.csv.

With historical returns and only long-only limits, the optimiser puts 50% in US stocks, 42% in gold and 7% in Saudi equities, and holds no bonds. That is simply the decade's three best performers. It is a textbook case of the main weakness of mean-variance optimisation: it is very sensitive to expected-return inputs and turns estimation error into concentrated bets. With policy bounds, the historical maximum-Sharpe portfolio is more sensible, but still sets gold and cash at their upper limits and leaves out emerging markets and REITs entirely.

With forward-looking assumptions, the picture reverses. Bonds are now expected to return only about 2 percentage points less than US stocks, with a third of the volatility, so the optimiser moves to the minimum equity weight allowed (40%). The proposed portfolio sits between the two views. Its forward expected return of 6.09% matches the forward optimum (6.08%), although at higher volatility (8.3% against 6.9%). At the proposal's volatility the forward policy frontier offers 6.43%, only 0.33 points more. On historical returns the gap is larger: the historical policy frontier offers 8.83% at the same risk, against the proposal's 7.49%. That gap is the cost of not loading up on the past decade's winners. We regard this robustness across two very different sets of return assumptions as more valuable than being optimal under either one.

## 4.1 Risk contributions and risk parity

**Figure 5. Capital weights versus contributions to portfolio volatility**




Source: Covariance of monthly returns, Oct 2016 to Sep 2026; author's calculations.

Capital weights can hide where the risk really sits. Equities and REITs make up 48% of the proposed portfolio's capital but 79% of its volatility. US stocks alone contribute 43%, and bonds about 16%. The risk parity portfolio, which excludes cash and is unlevered, spreads risk equally across the nine risky funds. To do so it holds about 46% in bonds and only 37% in equities, of which almost half is in the two small, volatile Gulf funds. Without leverage, its expected return of 6.09% is similar to the proposal's, at lower volatility (7.8%). However, it fell short in the backtest (6.2% a year), and it holds about 17% in Gulf equities, which conflicts with the client's human-capital constraint. We do not adopt it, but it is a useful cross-check: it confirms that the proposal's risk is still mainly equity risk, and it supports the decision to hold more bonds and TIPS than a conventional 60/40 portfolio.

## 5. Historical backtest

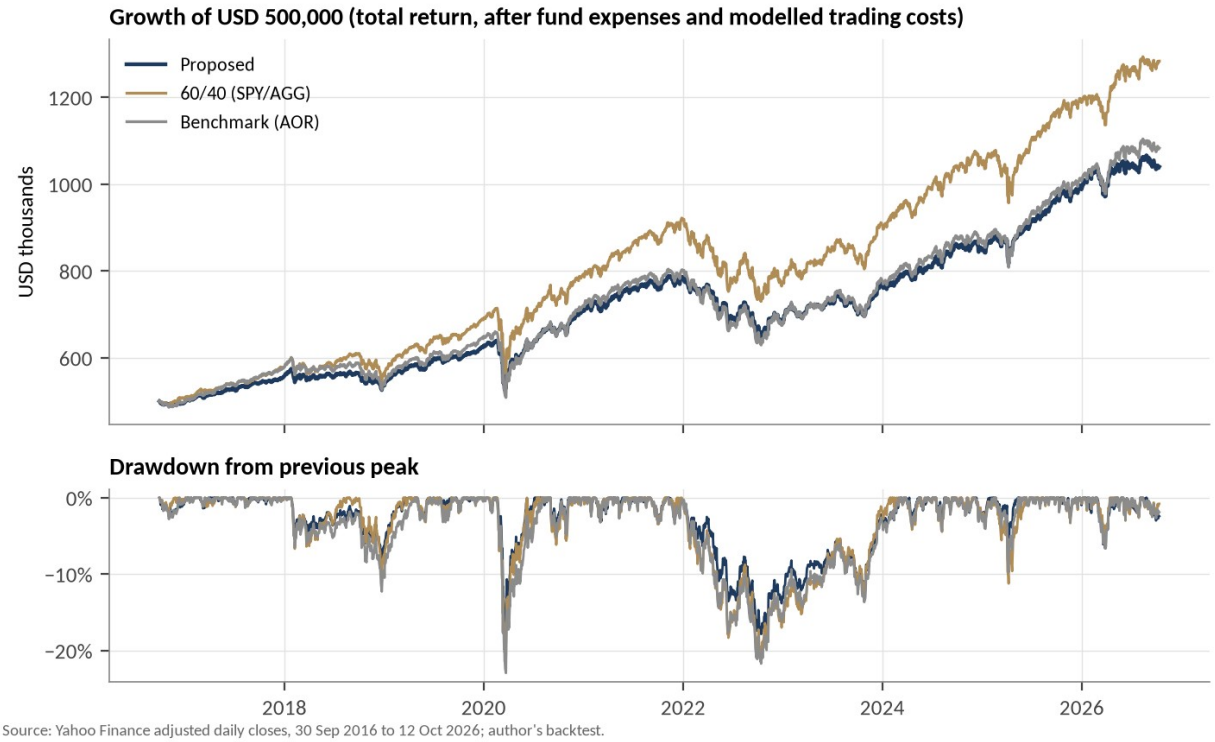
**Method.** The backtest simulates each portfolio day by day from 30 September 2016 to 12 October 2026, starting with USD 500,000. It uses adjusted closing prices, so dividends are reinvested and fund expenses are already deducted. At each quarter end, the proposed portfolio is rebalanced back to its target weights if any fund has drifted outside its tolerance band. The band is 3 percentage points or 25% of the target weight, whichever is smaller (for example, ±3 points for BND and ±0.75 points for KSA). The US 60/40 portfolio (SPY/AGG) is rebalanced every quarter, and AOR is held unchanged as a fund that rebalances itself. Every trade costs 5 basis points. The statistics below use the 120 complete months from October 2016 to September 2026. Sharpe and Sortino ratios use the 3-month Treasury bill rate as the risk-free rate. Maximum drawdown is measured on daily closes.

**Table 8. Backtest results, October 2016 to September 2026**

| Statistic | Proposed | US 60/40 (SPY/AGG) | Benchmark (AOR) |
|---|---|---|---|
| CAGR | 7.5% | 9.7% | 8.0% |
| Annualised volatility | 8.3% | 10.2% | 10.0% |
| Sharpe ratio | 0.62 | 0.72 | 0.57 |
| Sortino ratio | 0.92 | 1.10 | 0.84 |
| Maximum drawdown (daily) | −17.9% | −21.0% | −22.9% |
| Maximum drawdown (month-end) | −17.0% | −20.1% | −20.8% |
| Calmar ratio (CAGR / max drawdown) | 0.42 | 0.46 | 0.35 |
| Worst calendar year (2017-2025) | −13.0% (2022) | −16.0% (2022) | −15.6% (2022) |
| Best calendar year (2017-2025) | 16.6% (2025) | 21.8% (2019) | 18.9% (2019) |
| Worst month | −7.3% | −7.4% | −9.5% |
| Beta to S&P 500 | 0.49 | 0.65 | 0.62 |
| Value of USD 500,000 at 30 Sep 2026 | 1,034,019 | 1,265,325 | 1,074,538 |
| Rebalancing trades (10 years) | 10 | 40 | 0 |
| Average annual turnover | 9.0% | 12.6% | 0.0% |
| 2026 year to date (to 12 Oct 2026) | 5.3% | 8.0% | 7.6% |

Monthly statistics from 120 month-end returns. Returns are after fund expense ratios and modelled trading costs, before advisory fees and withholding tax.

**Figure 6. Growth of USD 500,000 and drawdowns**

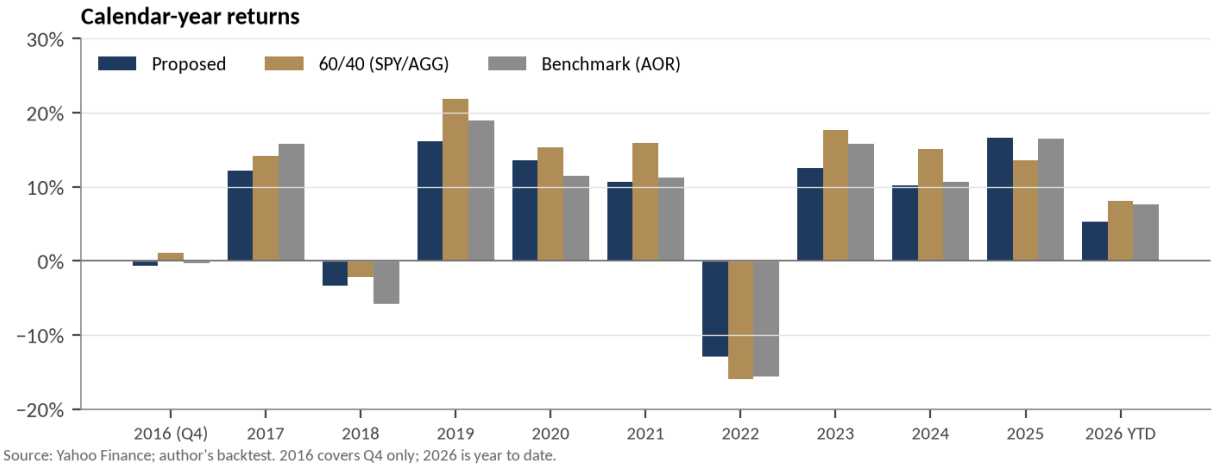



Source: Yahoo Finance adjusted daily closes, 30 Sep 2016 to 12 Oct 2026; author's backtest.

**Interpretation.** The proposed portfolio returned 7.5% a year, with 8.3% volatility and a Sharpe ratio of 0.62. It did slightly worse than the AOR benchmark on return (8.0%) but had lower volatility and a better Sharpe ratio (0.57), Sortino ratio and maximum drawdown. The US 60/40 portfolio did best on all return measures (9.7% a year, Sharpe 0.72). That result reflects one exceptional decade for US large-cap stocks: VTI compounded at 14.7% a year, against 9.7% for developed ex-US markets and 7.6% for emerging markets. Bonds, meanwhile, earned almost nothing in real terms. A portfolio that was concentrated in the US and had no inflation protection could not have been beaten over this particular period. We do not think it is wise to assume the same conditions for the next ten years, given today's valuations and yields (Section 2).

The proposal's strength is in protecting against losses, which is what this client needs most, given his 2022 experience. Its largest fall was −17.9%, from 9 Nov 2021 to 14 Oct 2022, and it had recovered by 7 Mar 2024. That is within the 20% tolerance in the IPS. The 60/40 portfolio fell −21.0% and AOR fell −22.9%, both in the 2020 COVID crash. The proposal's beta to the S&P 500 is 0.49, so on average it captures about half of the market's moves.

**Figure 7. Calendar-year returns**




Source: Yahoo Finance; author's backtest. 2016 covers Q4 only; 2026 is year to date.

## 5.1 Stress periods

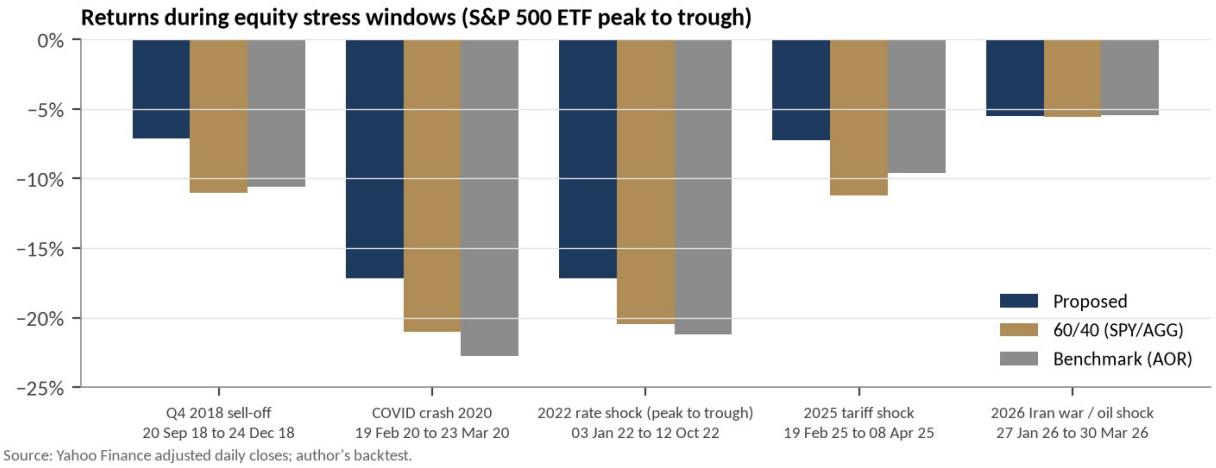
**Table 9. Returns in stress periods**

| Episode | Dates | Proposed | US 60/40 | AOR |
|---|---|---|---|---|
| Q4 2018 sell-off | 20 Sep 2018 to 24 Dec 2018 | −7.1% | −11.0% | −10.6% |
| COVID crash 2020 | 19 Feb 2020 to 23 Mar 2020 | −17.2% | −21.0% | −22.7% |
| Calendar 2020 | 31 Dec 2019 to 31 Dec 2020 | 13.6% | 15.3% | 11.4% |
| 2022 rate shock (peak to trough) | 3 Jan 2022 to 12 Oct 2022 | −17.2% | −20.4% | −21.2% |
| Calendar 2022 | 31 Dec 2021 to 31 Dec 2022 | −13.0% | −16.0% | −15.6% |
| 2025 tariff shock | 19 Feb 2025 to 8 Apr 2025 | −7.2% | −11.2% | −9.6% |
| 2026 Iran war / oil shock | 27 Jan 2026 to 30 Mar 2026 | −5.5% | −5.6% | −5.4% |

Peak-to-trough windows are defined by the S&P 500 ETF (SPY) high and low within each episode and applied to all portfolios.

In the **2020 COVID crash**, the proposal fell −17.2%, against −21.0% for the 60/40 and −22.7% for AOR. While equity funds fell 28%-42%, bonds, TIPS and gold lost only 1%-4% and T-bills gained, and the portfolio still gained 13.6% for the year. **2022** was the harder test for any balanced portfolio, because stocks and bonds fell together as the Fed raised rates quickly. The proposal lost −13.0% for the year, against −16.0% for the 60/40. It was helped by its T-bills, by gold, which ended the year roughly flat, and by the Gulf funds, which fell only 5%-6%. In the **2026 oil shock** (27 Jan 2026 to 30 Mar 2026), the three portfolios fell by similar amounts, about 5.5%. Gold, which had peaked just before, fell along with equities. Even good hedges do not work in every episode.

**Figure 8. Returns during equity stress windows**




Source: Yahoo Finance adjusted daily closes; author's backtest.

## 5.2 Sensitivity to the rebalancing rule

**Table 10. Effect of the rebalancing rule on the proposed portfolio**

| Rule | CAGR | Volatility | Sharpe | Max drawdown | Rebalancing events | Turnover p.a. |
|---|---|---|---|---|---|---|
| Quarterly check, tolerance bands (proposed) | 7.54% | 8.28% | 0.62 | −17.9% | 10 | 9.0% |
| Monthly calendar | 7.38% | 8.33% | 0.60 | −18.4% | 120 | 23.1% |
| Quarterly calendar | 7.45% | 8.31% | 0.61 | −18.0% | 40 | 14.5% |
| Annual calendar (December) | 7.40% | 8.22% | 0.61 | −17.8% | 10 | 7.3% |
| Never rebalance (buy and hold) | 8.38% | 9.35% | 0.64 | −19.5% | 0 | 0.0% |

The tolerance-band rule produced results at least as good as calendar rebalancing, with far fewer trades: 10 rebalancing events in ten years, against 120 for monthly rebalancing. Never rebalancing would have earned more (8.38% a year), but only because the equity weight drifted upward during a long bull market. Volatility and drawdown rose to 9.4% and −19.5%, close to the limits of the client's risk objective, and the risk would have kept rising the longer the drift continued. Rebalancing is a risk-control tool, not a way to add return.

# 6. Risk analysis

## 6.1 Value at Risk and Expected Shortfall

Value at Risk (VaR) is the loss that should be exceeded only with a given probability over a given period. Conditional VaR (CVaR, also called Expected Shortfall) is the average loss in the cases where VaR is exceeded. We estimate both in two ways. The historical method uses the portfolio's actual simulated returns. The parametric method assumes normally distributed returns with the same mean and volatility.

**Table 11. Value at Risk and CVaR for the proposed portfolio**

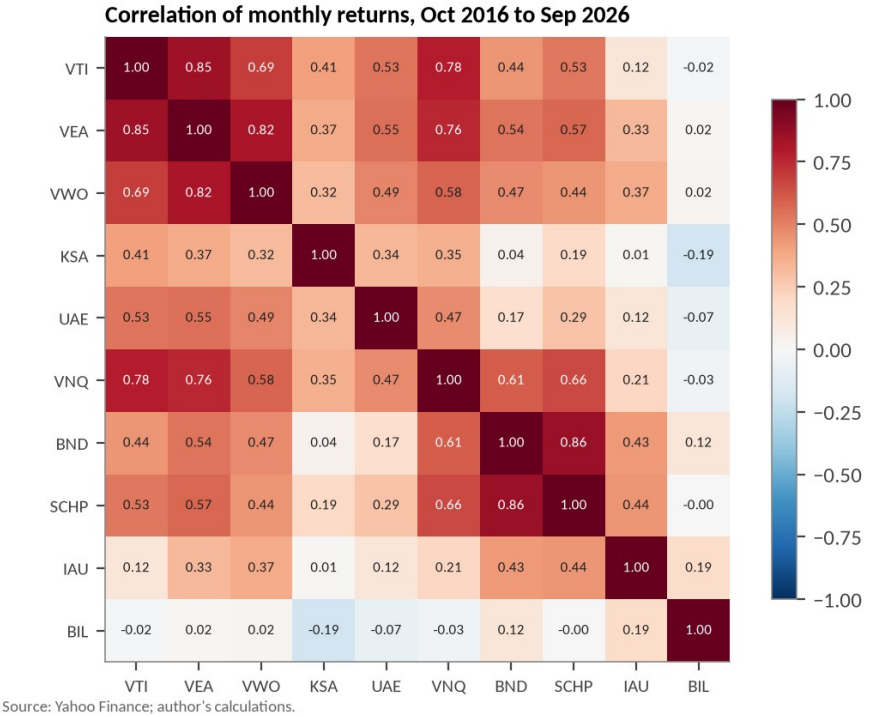
| Horizon, confidence | Hist. VaR | Hist. CVaR | Param. VaR | VaR (USD) | CVaR (USD) | Basis |
|---|---|---|---|---|---|---|
| 1 day, 95% | 0.78% | 1.29% | 0.87% | 3,899 | 6,442 | 2,520 daily returns |
| 1 day, 99% | 1.44% | 2.29% | 1.24% | 7,211 | 11,475 | 2,520 daily returns |
| 1 month, 95% | 3.21% | 5.18% | 3.30% | 16,047 | 25,902 | 120 monthly returns |
| 1 month, 99% | 6.13% | 6.88% | 4.92% | 30,651 | 34,400 | 120 monthly returns |
| 1 year, 95% | n/a | n/a | 7.61% | 38,027 (param.) | 55,428 (param.) | Forward assumptions |
| 1 year, 99% | n/a | n/a | 13.28% | 66,406 (param.) | 80,517 (param.) | Forward assumptions |

USD amounts for 1-day and 1-month horizons are historical estimates; 1-year amounts are parametric, using the forward expected return and the 10-year volatility.

On a normal day, losses should exceed 0.78% (about USD 3,899) only one day in 20. Over a year, under the forward assumptions, there is a 5% chance of losing more than 7.6%. That is inside the IPS limit of 10%. At the 99% level, historical VaR is clearly larger than parametric VaR (monthly: 6.13% against 4.92%). This is the familiar "fat tails" effect: the worst outcomes are worse than a normal distribution predicts. For that reason we rely on the stress tests and scenarios below, not on VaR alone.

## 6.2 Correlations

**Figure 9. Correlation matrix of monthly returns**

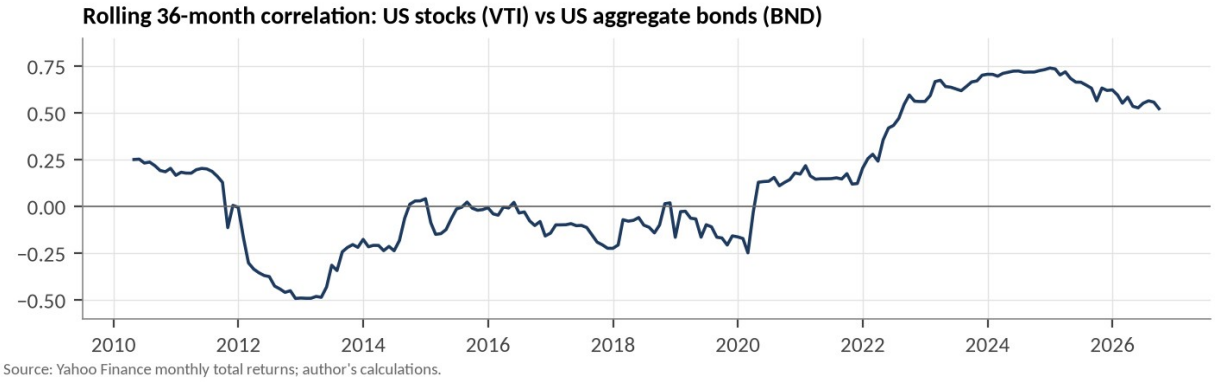



Source: Yahoo Finance; author's calculations.

Equity sleeves are highly correlated with one another (0.69-0.85 among VTI, VEA and VWO). The Gulf funds are much less correlated with them (0.32-0.55), which is what makes a small Gulf allocation useful. Gold has the lowest correlation with US equities of any risky asset (0.12). The key risk for a balanced portfolio is the

correlation between stocks and bonds. Figure 10 shows that it was negative for most of 2012-2020, then turned strongly positive after inflation returned in 2022. The 36-month figure was 0.52 at 30 Sep 2026. With inflation still above target, bonds cannot be relied on to offset equity losses in an inflationary sell-off. That is why the portfolio also holds TIPS, gold and T-bills.

**Figure 10. Rolling 36-month stock-bond correlation**




Source: Yahoo Finance monthly total returns; author's calculations.

## 6.3 Scenario analysis

**Table 12. Hypothetical scenario impacts**

| Hypothetical scenario (instantaneous) | Proposed | Proposed (USD) | 60/40 (VTI/BND) |
|---|---|---|---|
| Global equity bear market | −12.8% | −63,850 | −16.4% |
| Rates +150bp (bond sell-off) | −8.3% | −41,550 | −9.6% |
| Stagflation, Brent to $150 | −6.6% | −33,150 | −11.0% |
| Gulf escalation / Hormuz closure | −4.6% | −22,800 | −5.2% |
| Recession, Fed cuts 200bp | −6.8% | −33,950 | −8.8% |
| US dollar +10% | −3.1% | −15,600 | −1.4% |

Asset-level shocks are illustrative assumptions informed by past episodes and fund durations (about 6 years for BND and TIPS). They are listed in full in the Excel workbook (Scenarios sheet) and can be changed there.

The worst of the scenarios we tested is a global equity bear market with equities down 30% (US) to 35% (EM). In it, the portfolio would lose about 12.8%, or USD 63,850, partly offset by gains in Treasuries and gold. In a stagflation scenario with Brent at USD 150, the proposal would lose clearly less than a 60/40 portfolio (−6.6% against −11.0%), because TIPS, gold and Saudi equities would cushion the loss. A 10% rise in the dollar hurts the proposal slightly more than a US-only 60/40, because of its non-US equities and gold. For a client whose spending is in dirhams pegged to the dollar, this is a moderate and acceptable risk.

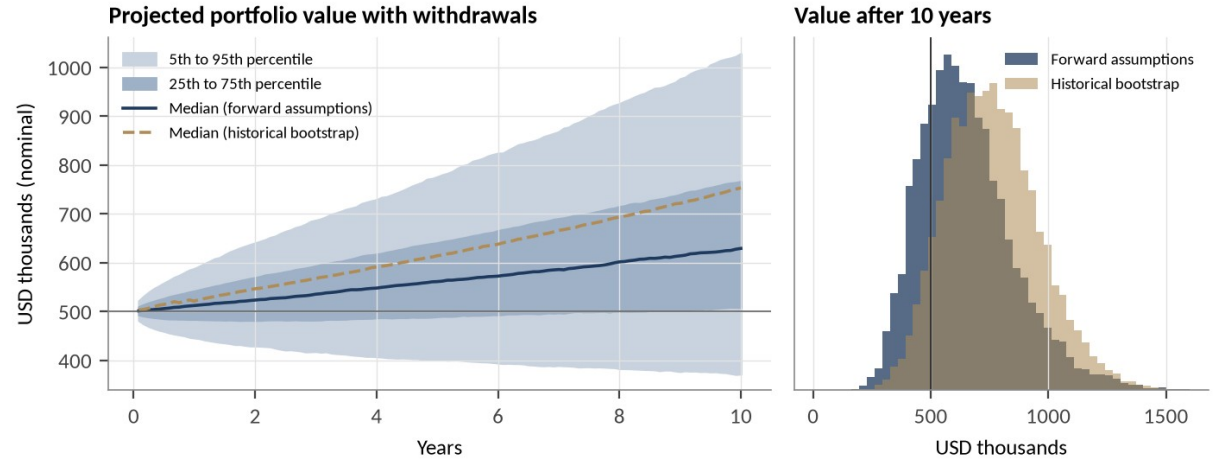
## 6.4 Monte Carlo projection and income sustainability

We simulated 10,000 ten-year paths of monthly returns. Each month the portfolio was rebalanced, a 0.50% annual fee was deducted, and USD 1,250 was withdrawn, with the withdrawal increased each year by the 2.33% breakeven inflation rate. That adds up to USD 166,746 withdrawn over ten years. We used two methods. The **forward-assumption simulation** draws returns from a multivariate normal distribution with the expected returns in Table 6 and the ten-year historical covariance matrix. The **historical bootstrap** resamples actual 12-month blocks of the past decade's monthly returns, which keeps fat tails and real correlation patterns, but carries forward the strong returns of that decade.

**Table 13. Monte Carlo results after 10 years, with withdrawals and fees**

| Portfolio value after 10 years (USD) | Forward assumptions | Historical bootstrap |
|---|---|---|
| 5th percentile | 371,798 | 463,234 |
| 25th percentile | 508,221 | 619,940 |
| Median | 628,818 | 752,824 |
| 75th percentile | 765,933 | 890,091 |
| 95th percentile | 1,027,713 | 1,114,327 |
| Median in today's dollars | 499,453 | 597,948 |
| 5th percentile in today's dollars | 295,310 | 367,935 |
| Probability nominal value ≥ USD 500,000 | 76% | 92% |
| Probability real value ≥ USD 500,000 | 50% | 73% |
| Probability value < USD 400,000 | 8% | 2% |
| Probability portfolio runs out within 10 years | 0% | 0% |

**Figure 11. Monte Carlo projection: percentile bands and distribution of ending values**




Source: Author's simulation, 10,000 paths; USD 15,000 a year withdrawn, indexed to inflation; 0.50% annual fee.

**Table 14. Income sustainability by withdrawal rate (inflation-indexed, after 0.50% fee)**

| Initial withdrawal rate | Lasts 30 years (forward) | Lasts 30 years (bootstrap) | Real capital kept at 10 years (forward) | Real capital kept at 10 years (bootstrap) |
|---|---|---|---|---|
| 2% | 100% | 100% | 64% | 85% |
| 3% | 98% | 100% | 50% | 72% |
| 4% | 83% | 97% | 35% | 57% |
| 5% | 51% | 82% | 23% | 41% |
| 6% | 22% | 51% | 12% | 25% |

At the planned 3% rate, the portfolio did not run out in any simulated ten-year path. Over a 30-year horizon it lasted in 98% of forward-assumption paths. The more demanding test is keeping purchasing power. Under the forward assumptions, the real value after ten years is about as likely to be above USD 500,000 as below it (50%). Under the historical bootstrap the probability is 73%. Withdrawals of 4% or more noticeably raise the risk of depleting the portfolio over a retirement-length horizon. Our recommendation is to keep the withdrawal at or below USD 15,000 a year, to review it each year against the portfolio's value, and to skip the inflation increase in any year after the portfolio has fallen more than 10%.

# 7. Implementation, costs, rebalancing and monitoring

## 7.1 Choice of fund domicile

For a non-US investor, where a fund is domiciled matters as much as its fee. US-listed ETFs have the longest histories, the tightest dealing spreads and the lowest headline fees. However, they expose a non-US investor to 30% withholding tax on dividends and to US estate tax on US-situs assets. Irish-domiciled UCITS ETFs pay 15% US withholding on US dividends at fund level, under the US-Ireland treaty. They are not US-situs assets for estate tax, and accumulating share classes reinvest income automatically. We recommend implementing the core sleeves through the UCITS equivalents in Table 15, and keeping the UAE country fund in US-listed form, as no comparable UCITS line was identified. Before any purchase, the bank should confirm the fees in each fund's Key Information Document.

**Table 15. Recommended UCITS implementation**

| Sleeve | Analysed | UCITS alternative | Ticker | ISIN | TER |
|---|---|---|---|---|---|
| US equity | VTI | iShares Core S&P 500 UCITS ETF USD (Acc) | CSPX | IE00B5BMR087 | 0.07% |
| Developed ex-US equity | VEA | Xtrackers MSCI World ex USA UCITS ETF 1C | EXUS | IE0006WW1TQ4 | 0.15% |
| Emerging markets equity | VWO | iShares Core MSCI Emerging Markets IMI UCITS ETF (Acc) | EIMI | IE00BKM4GZ66 | 0.18% |
| Saudi equity | KSA | iShares MSCI Saudi Arabia Capped UCITS ETF USD (Acc) | IKSA | IE00BYYR0489 | 0.60% |
| US aggregate bonds | BND | iShares US Aggregate Bond UCITS ETF (Dist) | IUAG | IE00B44CGS96 | 0.25% |
| US TIPS | SCHP | iShares USD TIPS UCITS ETF USD (Acc) | ITPS | IE00B1FZSC47 | 0.10% |
| Gold | IAU | iShares Physical Gold ETC | IGLN | IE00B4ND3602 | 0.12% |
| US REITs | VNQ | iShares US Property Yield UCITS ETF | IUSP | IE00B1FZSF77 | 0.40% |
| Cash / T-bills | BIL | iShares USD Treasury Bond 0-1yr UCITS ETF (Acc) | IB01 | IE00BGSF1X88 | 0.07% |

Total expense ratios from justETF fund profiles, retrieved 14 Oct 2026. UCITS funds track similar but not identical indices (for example the S&P 500 instead of the total US market).

Holding US-listed funds would cost an estimated 0.27% a year in withholding tax on equity and REIT dividends. If bond fund distributions were also taxed, the upper bound would be 0.85%. US estate-tax exposure could apply to almost the whole portfolio. The UCITS route roughly halves the withholding cost on US dividends and removes most of the estate-tax exposure, at the price of somewhat higher fund fees: a weighted TER of about 0.17%, against 0.09% for the US-listed funds.

## 7.2 Sharia-compliant variant

If the client chooses the Sharia-compliant option, interest-bearing instruments are replaced and the equity funds are switched to Sharia-screened indices. The screens exclude prohibited sectors and companies with high debt or interest income. US aggregate bonds and TIPS become a global sukuk fund. The T-bill reserve becomes an Islamic money-market fund or murabaha deposits with the bank. REITs are dropped, because many fail leverage screens, and their weight moves to Islamic world equities. Physically backed gold is generally accepted by Sharia scholars, but the client should confirm the specific product with the bank's Sharia board.

**Table 16. Sharia-compliant variant**

| Sleeve | Conventional | Sharia-compliant alternative | Weight | Fee |
|---|---|---|---|---|
| World equity (US and developed) | VTI, VEA, VNQ (38%) | iShares MSCI World Islamic UCITS ETF (ISWD/ISDW) | 38% | 0.30% |
| Emerging markets equity | VWO (5%) | iShares MSCI EM Islamic UCITS ETF (ISDE) | 5% | 0.35% |
| Gulf equity | KSA, UAE (5%) | KSA, UAE (Sharia screening of holdings required) | 5% | 0.75% / 0.59% |
| Fixed income | BND, SCHP (37%) | SP Funds Dow Jones Global Sukuk ETF (SPSK) | 37% | 0.50% |
| Gold | IAU (7%) | Physically backed gold ETC (e.g. IGLN), subject to Sharia review | 7% | 0.12% |
| Cash | BIL (8%) | Islamic money-market fund or murabaha deposit | 8% | varies |

ISWD and ISDE fees from justETF (the Yahoo Finance profile shows higher figures; confirm with the Key Information Document). SPSK fee from stockanalysis.com and Yahoo Finance.

Sukuk ETF data only start in late 2019, so we can compare the two versions only from January 2020 to September 2026 (81 months). Over that period the Sharia variant returned 7.3% a year with 9.1% volatility (Sharpe 0.50), against 7.7% and 9.4% (Sharpe 0.53) for the conventional proposal. Maximum drawdowns were almost identical (−17.8% and −17.9%). The sukuk fund held up better in 2022: the Sharia variant lost −9.0% that year, against −13.0%. On this evidence, the client can choose the Sharia-compliant version without a material change in expected risk or return. The main differences are a less diversified sukuk market and no explicit inflation-linked asset. In the backtest, BIL stands in for Islamic cash returns, and the Islamic world equity fund's London price, quoted in pence, was converted to US dollars.

## 7.3 Execution

1. Open the account and complete onboarding: KYC, source of wealth, CRS self-certification, W-8BEN for any US-listed funds, and a signed IPS.
2. In week one, invest the defensive sleeves in full: bonds, TIPS, T-bills and gold (52% of the portfolio), together with the REIT sleeve (3%).
3. Invest the equity sleeves in three equal tranches over about six weeks. Historically, investing a lump sum at once has more often done better, so this short phase-in is a behavioural choice. It reduces the regret of investing everything just before a fall, which matters for this client given his 2022 experience.
4. Trade during liquid hours: European hours for UCITS funds and the US session for US-listed funds. Use limit orders close to the indicative value, and avoid the first and last 15 minutes of trading.

## 7.4 Costs

**Table 17. Estimated costs**

| Cost item | % of assets a year | USD a year |
|---|---|---|
| Fund expenses (weighted): US-listed / UCITS route | 0.09% / 0.17% | 458 / 859 |
| Trading costs (spread and commission), initial investment | about 0.05%-0.10% one-off | 250-500 one-off |
| Rebalancing (about 9% turnover a year at 5 bp) | below 0.01% | below 50 |
| Advisory and custody fee (assumption used in projections) | 0.50% | 2,500 |
| Withholding tax on dividends, US-listed funds (estimate) | 0.27%-0.85% | 1,338-4,242 |
| **Indicative all-in annual cost, UCITS route (fees, fund expenses, residual withholding)** | **about 0.8%** | **about 4,000** |

## 7.5 Rebalancing policy

- **Review frequency.** Weights are checked at each quarter end, and also within five business days of any month in which global equities move by more than 10%.
- **Tolerance bands.** Each fund may drift by 3 percentage points or 25% of its target weight, whichever is smaller. If any fund breaches its band, the whole portfolio is rebalanced to target. Asset-class totals must also stay within the IPS ranges in Table 3.
- **Use cash flows first.** Withdrawals are paid from the T-bill sleeve and from distributions. New money and distributions go to the most underweight sleeve before any sales are made.
- **Refill the income reserve.** If the T-bill sleeve falls below 5%, it is topped up at the next review from whichever sleeve is most overweight. Equities are never sold to fund withdrawals when they are more than 15% below their previous peak, unless the reserve is exhausted.
- **Discipline.** Rebalancing is rule-based and not changed on market views, which protects the client from selling at the bottom, as happened in 2022.

## 7.6 Monitoring and review

The client will receive a quarterly report showing performance against the policy benchmark, AOR and a 60/40 portfolio, current weights against the bands, income paid, realised volatility and drawdown, and any rebalancing trades. The IPS will be reviewed each year, and also after any material change in circumstances,

such as job loss or relocation, a change in tax residence, an inheritance, or a decision on the Sharia-compliant variant. The capital market assumptions will be refreshed each year from current yields. A formal review is triggered automatically if any of the following occurs:

- the portfolio falls more than 15% from its high (an early warning before the 20% tolerance);
- realised 12-month volatility rises above 12%;
- a fund's tracking difference exceeds 0.5% a year, or a fund changes its index, fee or structure;
- the 10-year real yield or the Fed policy rate moves by more than 1 percentage point from the levels assumed here;
- there are material changes in US withholding or estate-tax rules for non-residents, or in sanctions affecting payments to Syria.

# 8. Disclaimers

This document is a case study prepared by Yamen Agha for educational and professional-portfolio purposes. The client, Mr. Omar Haddad, is fictional, and any resemblance to a real person is coincidental. This document is not an offer or solicitation to buy or sell any security, and it is not personalised investment, tax, legal or Sharia advice.

Market data come from public sources believed to be reliable (Appendix A), but are not guaranteed to be accurate or complete. Data are as of the dates shown and will change.

Past performance, including simulated (backtested) performance, is not a reliable indicator of future results. Backtests are hypothetical. They benefit from hindsight in the choice of funds and rules, assume trades at closing prices, and do not reflect advisory fees, taxes or the effect of real-world behaviour.

Capital market assumptions, Monte Carlo projections and scenarios are estimates based on stated assumptions. Actual outcomes may differ materially, and losses larger than those shown are possible. The value of investments can go down as well as up, and investors may get back less than they invested.

Exchange-traded funds carry market, liquidity, tracking and counterparty risks. Single-country funds such as KSA and UAE are concentrated and may be more volatile. Gold pays no income. TIPS and bonds can fall in value when real or nominal yields rise.

Tax and estate rules for non-US persons are complex and depend on individual circumstances. They should be confirmed with qualified advisers in each relevant jurisdiction before investing. The Sharia compliance of any product should be confirmed by a qualified Sharia supervisory board.

# Appendix A. Data sources and as-of dates

**Table 18. Data sources (all retrieved 14 October 2026)**

| ID | Data item | As of | Source |
|---|---|---|---|
| S1 | Daily adjusted ETF closes (dividends reinvested) for VTI VEA VWO KSA UAE VNQ BND SCHP IAU BIL SPY AGG AOR SPUS HLAL SPSK | 2026-10-12 (latest close available from the feed at retrieval; 13 Oct ETF bars not yet published) | Yahoo Finance via yfinance 1.7.0 |
| S2 | US Treasury daily par yield curve (nominal) | 2026-10-13 | US Department of the Treasury |
| S3 | US Treasury daily real (TIPS) par yield curve | 2026-10-13 | US Department of the Treasury |
| S4 | FRED series (DGS3MO DGS2 DGS10 DGS30 DFII10 T10YIE DFF DFEDTARU DFEDTARL CPIAUCSL CPILFESL UNRATE DTWEXBGS DCOILBRENTEU BAMLC0A0CM BAMLH0A0HYM2) | various (daily series to 2026-10-09/13; CPI Aug-2026; unemployment Sep-2026) | Federal Reserve Bank of St. Louis (FRED) |
| S5 | FOMC decision: target range raised 25bp to 3.75%-4.00% (12-0 vote); SEP median PCE inflation 3.7% (2026) and 2.3% (2027); GDP 2.3% (2026) | 2026-09-16 | Federal Reserve Board: FOMC statement and Chair's press conference transcript |
| S6 | US CPI August 2026: headline +3.4% y/y; core +2.4% y/y; energy +16.3% y/y; next release (September CPI) 14 Oct 2026 08:30 ET | 2026-09-11 | US Bureau of Labor Statistics CPI news release |
| S7 | Brent futures settle $105.69 (13 Oct) after $106.65 on 12 Oct (highest since 15 Sep); WTI $94.14; Houthi attacks on Saudi Arabia and US-Russia diesel arrangement | 2026-10-13 | Reuters (via financialnewsletter.com) and Economic Times |
| S8 | US equity close: S&P 500 7805.45 (-0.02%); Dow 51670.10; Nasdaq 27204.38; 10Y 5.21%; VIX 15.27; dollar index futures 102.05 | 2026-10-13 | Associated Press market report; Dow Jones / Morningstar Data Talk |
| S9 | 10Y Treasury 5.216% (Tradeweb 3pm); 52-week high 5.310% on 5 Oct 2026; +1.06pp year to date | 2026-10-13 | Dow Jones Newswires via Morningstar |
| S10 | Gold: record about $5590/oz intraday on 28 Jan 2026; about 25% below record in mid-October; COMEX front-month about $4180 on 13 Oct (Yahoo GC=F; see data_asof.json) | 2026-10-13 | Yahoo Finance (GC=F) and metalcharts.org summary |
| S11 | IMF World Economic Outlook October 2026: world growth 3.1% (2026) and 3.4% (2027); global headline inflation 4.7% (2026) and 4.1% (2027); Middle East and Central Asia -0.3% in 2026; oil price assumption $95.74 (2026) | 2026-10-13 | IMF WEO October 2026 (via AP and CNBC coverage) |
| S12 | Gulf markets: Tadawul All Share Index closed 10360.74 after Riyadh airport attack; index had erased its 2026 gains; US-Israeli war with Iran in its eighth month | 2026-10-11 | Reuters via Zawya; 60 Arabia |
| S13 | ETF expense ratios dividend yields and AUM | 2026-10-14 | stockanalysis.com ETF pages cross-checked with Yahoo Finance fund profile (netExpenseRatio) |
| S14 | UCITS (Irish-domiciled) alternatives and TERs | 2026-10-14 | justETF fund profiles |
| S15 | US estate tax for non-resident non-citizens: US-situs assets include stock of US corporations; Form 706-NA filing threshold $60000 (not indexed) | 2025-09 (form instructions) | Internal Revenue Service |
| S16 | US income tax treaty list (no treaty with UAE or Saudi Arabia; statutory 30% withholding on US-source dividends applies to non-treaty residents) | 2025-09-26 (Table 3) | Internal Revenue Service |

Full URLs are listed in data/sources.csv. FRED series are as of their own latest observation (data/data_asof.json). ETF prices for 13 October had not yet been published by the price feed at retrieval, so the latest ETF close used is 12 October 2026.

# Appendix B. Methodology and formulas

**Returns.** Monthly simple returns from month-end adjusted closing prices: r = P(t) / P(t-1) - 1. Annualised arithmetic mean = 12 x average monthly return. CAGR = (product of (1 + r))^(12/n) - 1.

**Volatility.** Sample standard deviation of monthly returns x √12.

**Sharpe ratio.** (Average monthly return - average monthly risk-free rate) x 12 / annualised volatility. The risk-free rate is the 3-month Treasury constant-maturity yield (FRED DGS3MO) at the previous month end, divided by 12.

**Sortino ratio.** Annualised average excess return divided by annualised downside deviation, where downside deviation = √(average of $\min(r - rf, 0)^2$) x √12.

**Maximum drawdown.** The largest percentage fall from a running peak in portfolio value, measured on daily closes, and also on month-end values for comparison with the Excel workbook.

**Mean-variance optimisation.** Minimise $w'\Sigma w$ subject to $w'\mu$ = target, $\Sigma w = 1$ and the weight bounds, solved with SLSQP (scipy) over 40 target returns. The maximum-Sharpe portfolio maximises $(w'\mu - rf) / \sqrt{(w'\Sigma w)}$.

**Risk parity.** Equal risk contribution: $w_i(\Sigma w)_i / (w'\Sigma w) = 1/N$ for the nine risky funds, with weights summing to 1 and no leverage.

**Risk contribution.** $RC_i = w_i(\Sigma w)_i / (w'\Sigma w)$. The contributions sum to 100% of portfolio variance.

**VaR and CVaR.** Historical VaR = minus the 5th (1st) percentile of returns, using linear interpolation. CVaR = minus the average of returns at or below that percentile. Parametric VaR = $-(\mu - z\sigma)$, with $z = 1.645$ (2.326). Parametric CVaR = $-(\mu - \sigma\varphi(z) / (1 - \alpha))$.

**Backtest.** Daily simulation with quarter-end checks of the tolerance bands, rebalancing to target when breached, a 5 bp cost on traded value, and dividends reinvested through adjusted prices.

**Monte Carlo.** 10,000 paths, random seed 20261014, monthly steps, monthly rebalancing. Forward case: multivariate normal returns with the Table 6 means and the 10-year sample covariance. Bootstrap case: random 12-month blocks of historical monthly returns. The fee is deducted monthly; the withdrawal of USD 1,250 a month is increased annually by 2.33%.

All calculations can be reproduced by running scripts/run_all.sh. The Excel workbook rebuilds the main statistics with live formulas and reconciles them with the Python results; all reconciliation checks pass after LibreOffice recalculation (excel/recalc_verification.txt).

## Appendix C. Asset statistics

**Table 19. Ten-year statistics by fund, October 2016 to September 2026**

| Ticker | Asset | CAGR | Volatility | Sharpe | Max drawdown (month-end) | First price in dataset |
|---|---|---|---|---|---|---|
| VTI | US total stock market | 14.7% | 15.8% | 0.80 | −24.8% | 3 Jan 2006 |
| VEA | Developed ex-US equity | 9.7% | 15.5% | 0.52 | −28.1% | 26 Jul 2007 |
| VWO | Emerging markets equity | 7.6% | 15.5% | 0.39 | −31.9% | 3 Jan 2006 |
| KSA | Saudi Arabia equity | 8.4% | 18.0% | 0.40 | −30.5% | 17 Sep 2015 |
| UAE | UAE equity | 5.9% | 17.6% | 0.27 | −42.4% | 1 May 2014 |
| VNQ | US REITs | 4.4% | 17.7% | 0.19 | −32.8% | 3 Jan 2006 |
| BND | US aggregate bonds | 1.2% | 5.2% | −0.22 | −17.3% | 10 Apr 2007 |
| SCHP | US TIPS | 2.1% | 5.0% | −0.06 | −13.8% | 5 Aug 2010 |
| IAU | Gold | 11.9% | 15.5% | 0.65 | −23.8% | 3 Jan 2006 |
| BIL | 1-3 month T-bills | 2.3% | 0.6% | −0.38 | −0.2% | 30 May 2007 |